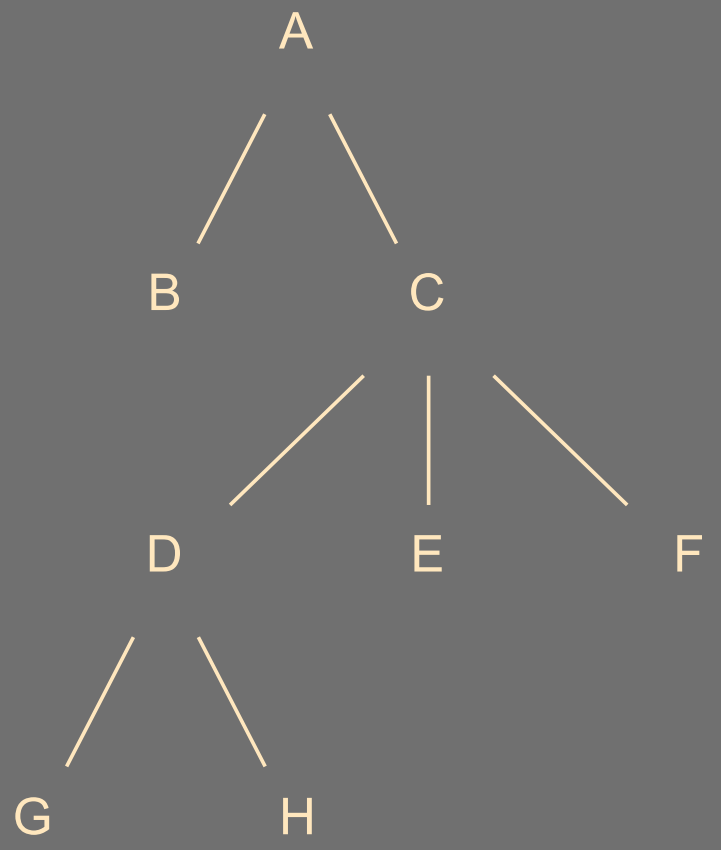
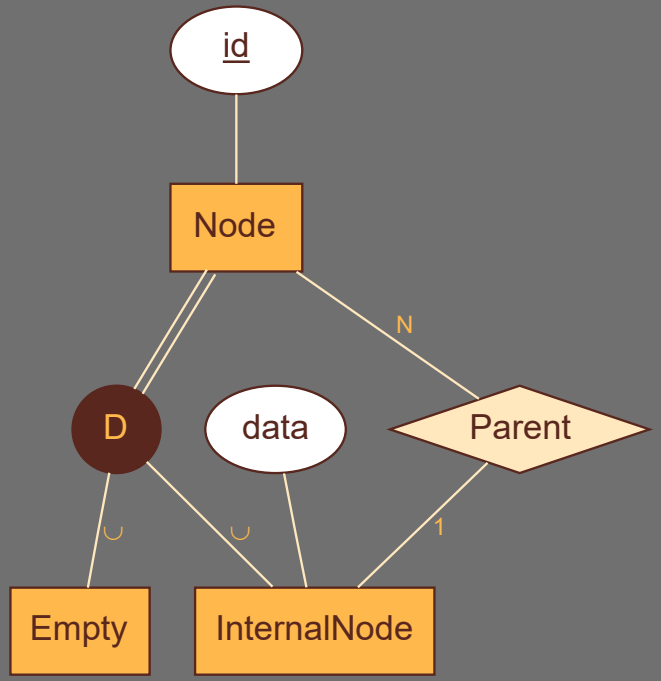
Tree-based/Hierarchical Design

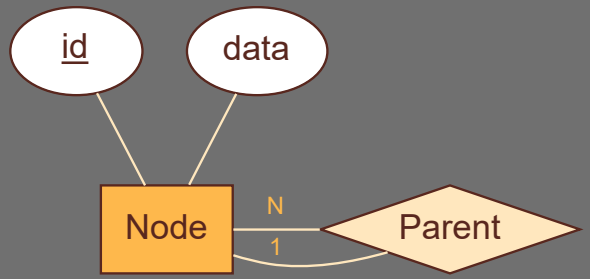
What are trees?



Composite / Structured tree / Adjacency list



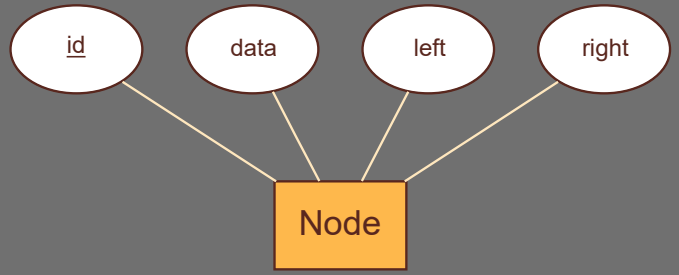
Simplified Composite



Node

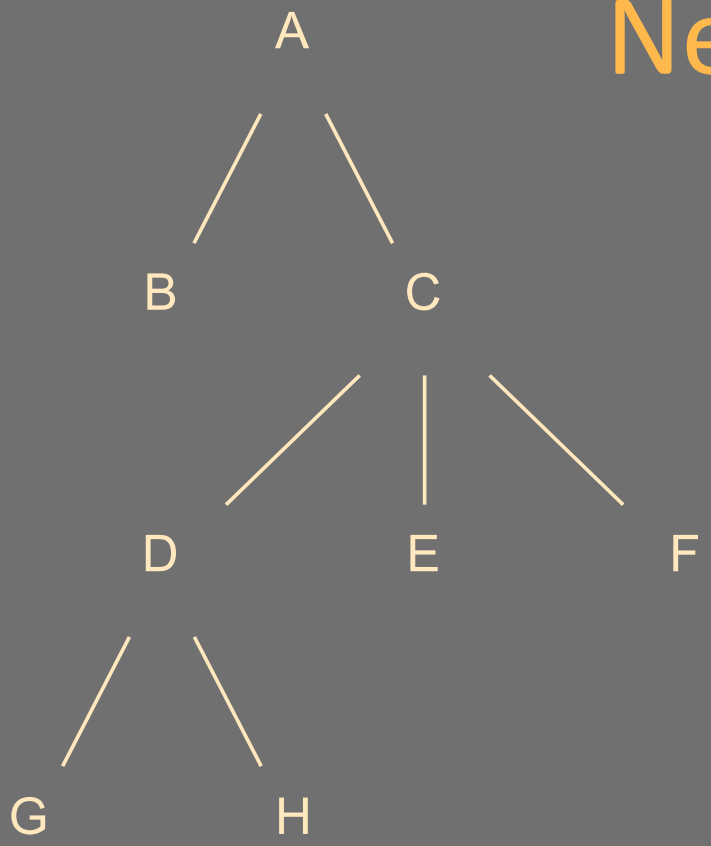
id	data	parent
A	This	NULL
B	is	A
C	just	A
D	silly	C
E	filler	C
F	data	C
G	for	D
H	example	D

Nested set



Simple data, but how can this represent a tree?

Nested Set





Tree representations

Name(s)	Attributes	Efficient operations
Composite, Structured Tree, Adjacency List	ID, parent ID	Insert, Delete, parent, children
Nested Set, Modified Preorder Tree Traversal	Left, Right (Integers)	Level, ancestors, descendants
Nested Intervals	Left, Right (Floats)	Insert, Delete, Level, ancestors, descendants
Bridge Table, Closure Table	• ID, parent ID • ID, descendant ID, level (transitive closure)	Level, parent, children, ancestors, descendants
Flat Table	ID, parent ID, level, rank	Level, ancestors, descendants
Lineage, Materialized Path, Path Enumeration	Lineage (like a file system path)	Parent, children, descendants
Multiple Lineage	Level 1 group, Level 2 group, Level 3 group, …	Insert, Delete, Level, parent, children, ancestors, descendants